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(Fall 2023)

Quantum Mechanics

Solution

(i)

What is meant by operators and state vectors in quantum mechanics? Also explain dimensionality and basis?

Operators and state vector:-

In quantum mechanics, physical state of a system is represented by elements of Hilbert space H called state vectors. It is independent of basis in which it is expanded. ket vectors represents the system completely.

If $|\psi\rangle$ and $|\phi\rangle$ belong to same vector space, products of the type $|\psi\rangle|\phi\rangle$ and $\langle\psi|\langle\phi|$ are not allowed because these are neither ket nor Bra. For each linear operator $\hat{A}$, we can write:

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$$\hat{A} = I\hat{A}I = \left(\sum_{i=1}^{\infty} |\psi_i\rangle\langle\psi_i|\right)\hat{A}\left(\sum_{j=1}^{\infty}|\psi_j\rangle\langle\psi_j|\right)$$

$$= \sum_{i=1}^{\infty} A_{ij}|\psi_i\rangle\langle\psi_j|$$

where "A_{ij}" are the matrix elements of operator $\hat{A}$.

$$A_{ij} = \langle\psi_i|\hat{A}\psi_j|$$

Dimensionality and basis:-

A set of linearly independent vectors $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}$ in a vector space V such that every vector of V can be represented/ expressed as linear combination of vectors $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}$ is called a basis for vector space V.

The number of basis vector of a vector space V is called dimension of that space.

-(ii)-

What is meant by discrete and continuous spectrum? Explain it by considering example of general potential.

For bound states, set of eigen

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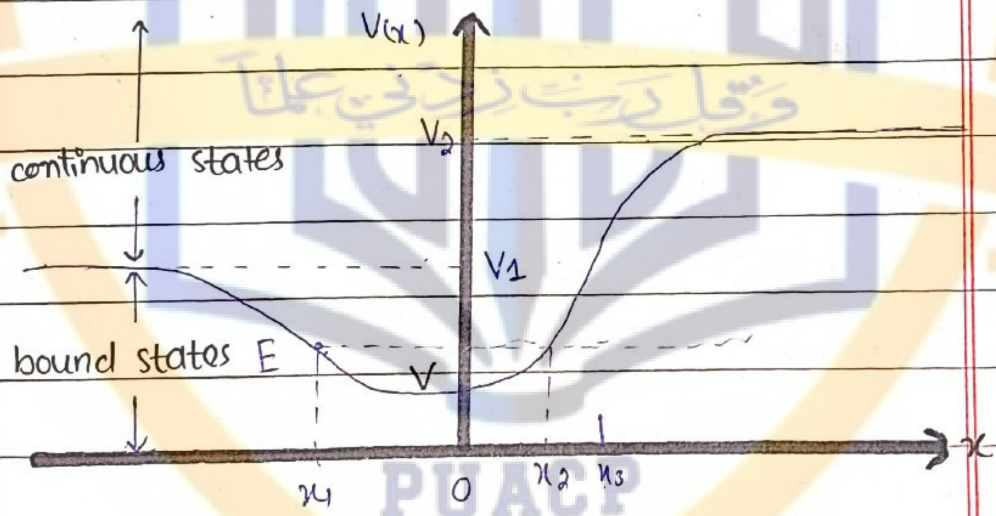
values and functions are discrete sets. The spectrum of those states is called discrete spectrum. For unbound states, set of eigen function and eigen values are continuous sets and spectra of such states is called continuous spectra. In the figure, motion of particle is bounded between points "x_1" and "x_2" when energy of particle lies between V_0 and V_1.



The motion of particle is unbounded for energy ranges $V_1 < E < V_2$ and $E > V_2$. The states corresponding to this energy range are called unbound energy states. The spectrum of bound states is discrete and non-degenrate. Unbound states cann't be normalized.

(iii)

State any three postulates of quantum mechanics.

i- For a system consisting of particle moving in a conservative force, there is an associated complex wave function $\psi(x,y,z,t)$, where x, y, z are space coordinates and "t" is time. This function enables us to obtain a description of the behaviour of system consistent with uncertainty principle.

ii- The wave function $\psi(x,y,z,t)$ and its partial derivatives $\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}, \frac{\partial\psi}{\partial z}$ must be finite, continuous and single valued for all values of "x, y, z and t".

iii- The average or expectation value of an observable quantity "A" with which an operator $\hat{A}$ is associated is defined by:

$$\hat{A} = \frac{\int_{-\infty}^{+\infty} \psi^* \hat{A} \psi \, dV}{\int_{-\infty}^{+\infty} \psi^* \psi \, dV} \qquad \because \int_{-\infty}^{+\infty} \psi^* \psi \, dV = 1$$

$$\Rightarrow \hat{A} = \int_{-\infty}^{+\infty} \psi^* \hat{A} \psi \, dV.$$

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(iv)

If x and p are position and momentum variables and

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

show that $[\hat{A}, \hat{A}^\dagger] = 1$.

since, $$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

$$\Rightarrow \hat{A}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p})$$

$$[\hat{A}, \hat{A}^\dagger] = \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} \Rightarrow \frac{1}{2m\hbar\omega}\{(m\omega\hat{x} + i\hat{p})(m\omega\hat{x} - i\hat{p})\}$$

$$- \frac{1}{2m\hbar\omega}\{(m\omega\hat{x} - i\hat{p})(m\omega\hat{x} + i\hat{p})\}$$

$$= \frac{1}{2m\hbar\omega}\{m^2\omega^2\hat{x}^2 + \hat{p}^2 + (-im\omega\hat{x}\hat{p}) + i\hat{p}m\omega\hat{x}\}$$

$$- \frac{1}{2m\hbar\omega}\{m^2\omega^2\hat{x}^2 + im\omega\hat{x}\hat{p} - i\hat{p}m\omega\hat{x} + \hat{p}^2\}$$

$$= \frac{1}{2m\hbar\omega}\left[\cancel{m^2\omega^2\hat{x}^2} + \cancel{\hat{p}^2} - im\omega\hat{x}\hat{p} + im\omega\hat{p}\hat{x} - \cancel{m^2\omega^2\hat{x}^2} - im\omega\hat{x}\hat{p} + im\omega\hat{p}\hat{x} - \cancel{\hat{p}^2}\right]$$

$$= \frac{-1}{2m\hbar\omega}\left[im\omega\hat{x}\hat{p} - im\omega\hat{p}\hat{x} + im\omega\hat{x}\hat{p} - im\omega\hat{p}\hat{x}\right]$$

$$= \frac{-1}{2m\hbar\omega}\{im\omega[\hat{x}, \hat{p}] + im\omega[\hat{x}, \hat{p}]\}$$

$\hat{p}] = i\hbar$

$$= \frac{-1}{\cancel{2}m\hbar\omega}\cancel{2}(im\omega)([\hat{x}, \hat{p}]) \Rightarrow \frac{(-i)(i\cancel{\hbar})}{\cancel{\hbar}}$$

$$\Rightarrow [\hat{A}, \hat{A}^\dagger] = -i^2$$

$$\Rightarrow [\hat{A}, \hat{A}^\dagger] = 1$$ Hence proved.

-{ V }-

Find out classical amplitude of particle undergoing SHM if its energy is $\hbar\omega/2$.

For classical harmonic oscillator,

$$x = A_n \cos\omega t \quad \text{and} \quad p = m\frac{dx}{dt}$$

$$\Rightarrow P = m\frac{d}{dt}(A_n \cos\omega t)$$

$$P = -mA_n \sin\omega t \cdot \omega$$

Since energy is:

$$E_n = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$\Rightarrow E_n = \frac{m^{\cancel{2}}\omega^2 A_n^2 \sin^2\omega t}{2\cancel{m}} + \frac{1}{2}m\omega^2 A_n^2 \cos^2\omega t$$

$$= \frac{m\omega^2 A_n^2}{2}\{\sin^2\omega t + \cos^2\omega t\}$$

$$\Rightarrow E_n = \frac{m\omega^2 A_n^2}{2}(1) \quad \Rightarrow \quad A_n^2 = \frac{2E_n}{m\omega^2}$$

$$\Rightarrow A_n = \sqrt{\frac{2E_n}{m\omega^2}}$$

Which is classical amplitude of particle.

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(Vi)

A particle is in state given by wave function, $\psi(x,t) = e^{-ax^2-bt}$ where, a and b are constants. Show that is it a stationary state? Find expectation values of position and momentum operators.

Since, $\psi(x,t) = e^{-ax^2-bt}$

The probability density is given by:

$$P = \psi^*(x,t)\,\psi(x,t) = |\psi(x,t)|^2$$

$$P = e^{-2ax^2-bt}$$

Since probability density depends upon time. So, given state is not stationary state.

- Expectation value of position operator is given as:

$$\langle\hat{x}\rangle = \int_{-\infty}^{+\infty} \psi^*(x,t)\,\hat{x}\,\psi(x,t)\,dx$$

$$\Rightarrow \langle\hat{x}\rangle = \int_{-\infty}^{+\infty} e^{-ax^2-bt} \cdot x\, e^{-ax^2-bt} \cdot dx$$

$$\langle\hat{x}\rangle = \int_{-\infty}^{+\infty} x\, e^{-2ax^2-2bt} \cdot dx \Rightarrow \langle\hat{x}\rangle = 0.$$

- Expectation value of momentum operator is given as:

$$\langle\hat{p}_x\rangle = \int_{-\infty}^{+\infty} e^{-ax^2-bt}\, \hat{p}_x\, e^{-ax^2-bt} \cdot dx$$

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$$\langle \hat{p}_x \rangle = \int_{-\infty}^{+\infty} e^{-ax^2-bt}\left(-i\hbar\frac{\partial}{\partial x}\right)(e^{-ax^2-bt}) \cdot dx$$

$$\langle \hat{p}_x \rangle = -i\hbar \int_{-\infty}^{+\infty} e^{-ax^2-bt}(-2ax)e^{-ax^2-bt}\, dx$$

$$\langle \hat{p}_x \rangle = 2axi\hbar \int_{-\infty}^{+\infty} e^{-ax^2-bt} \cdot e^{-ax^2-bt} \cdot dx$$

$\because \int_{-\infty}^{+\infty} (\underset{\text{function}}{\text{even}} \times \underset{\text{function}}{\text{odd}})dx = 0$

$$\Rightarrow \langle \hat{p}_x \rangle = 0.$$

(vii)

Show that if an operator $\hat{A}$ has an eigen value $a \neq a^*$, then $\hat{A}$ is not Hermition.

Let ψ be eigen function of operator $\hat{A}$ (not Hermition) corresponding to eigen value, then:

$$\hat{A}\psi = a\psi \rightarrow (1)$$

Taking complex conjugate:-

$$\Rightarrow (\hat{A}\psi)^* = (a\psi)^* \rightarrow (2)$$

Multiplying eq(1) with ψ^* and integrating:

$$\Rightarrow \int_{-\infty}^{+\infty} \psi^* \hat{A}\psi dx = \int_{-\infty}^{+\infty} \psi^* a\psi dx$$

$$\Rightarrow \int_{-\infty}^{+\infty} \psi^* \hat{A}\psi dx = a\int_{-\infty}^{+\infty} \psi^* \psi dx \rightarrow (3)$$

Multiplying eq(2) with "ψ" and integrating:

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$$\Rightarrow \int_{-\infty}^{+\infty} (A\psi)^* \psi \, dx = \int_{-\infty}^{+\infty} \psi^* a^* \psi \, dx$$

$$\Rightarrow \int_{-\infty}^{+\infty} \psi^* A^\dagger \psi \, dx = a^* \int_{-\infty}^{+\infty} \psi^* \psi \, dx \rightarrow (4)$$

As $\hat{A}$ is not Hermition operator i.e. $a \neq a^*$.
So from eq (3) and (4), we get:

$$\int_{-\infty}^{+\infty} \psi^* A^\dagger \psi \, dx \neq \int_{-\infty}^{+\infty} \psi^* \hat{A} \psi \, dx$$

which shows that $\hat{A}$ is not Hermition operator.

(viii)

Given that in spherical polar coordinates $L_z = -i\hbar \frac{\partial}{\partial \Phi}$, show that: $[L_z, \sin\varphi] = -i\hbar \cos\varphi$

Since $L_z = -i\hbar \frac{\partial}{\partial \Phi}$

$$[L_z, \sin\Phi]\psi = L_z \sin\Phi \psi - \sin\Phi L_z \psi$$

$$= -i\hbar \frac{\partial}{\partial \varphi} (\sin\Phi)\psi - \sin\Phi \cdot \left(-i\hbar \frac{\partial \psi}{\partial \Phi}\right)$$

$$= \cancel{-i\hbar \sin\varphi \frac{\partial \psi}{\partial \Phi}} - i\hbar \cos\Phi \, \varphi + \cancel{i\hbar \sin\Phi \frac{\partial \psi}{\partial \Phi}}$$

$$\Rightarrow [L_z, \sin\Phi] = -i\hbar \cos\Phi$$

Hence proved.

-(ix)-

The operator for z-component of angular momentum is $L_z = -i\hbar\frac{\partial}{\partial\phi}$. Determine whether or not $\sin m\phi$ is its eigen function?

since, $\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}(\sin m\phi) \Rightarrow -i\hbar m\cos m\phi$$

This shows that $\sin m\phi$ is not an eigen function of $\hat{L}_z$. However, $e^{im\phi}$ is an eigen function of $\hat{L}_z$ corresponding to eigen value $m\hbar$.

-(X)-

State Results of stern Garlach experiment.

i- When an experiment was performed with silver, a fairy sharp line was traced on plate when magnetic field was off. On establishing a highly inhomogenous magnetic field, double trace was obtained on either side of original line.

ii- When experiment was performed with Cu, Au, H, Li, Na and K, then a double trace was found with application of non-homogeneous magnetic field.

iii. No effect was found for Zn, Cd, Hg, Sn and Pb.

LONG QUESTIONS

(I)

State and prove generalized uncertainty Relation (uncertainty principle for operators).

Given two operators $\hat{A}$ and $\hat{B}$ such that $[\hat{A},\hat{B}] = i\hat{C}$. The uncertainties $\Delta\hat{A}$ and $\Delta\hat{B}$ are related by in an orbitrary state as:

$$\Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2}|\langle\hat{C}\rangle|$$

Proof:-

since $(\Delta\hat{A})^2 = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2$

put $\hat{E} = \hat{A} - \langle\hat{A}\rangle$ and $\hat{F} = \hat{B} - \langle\hat{B}\rangle$

and consider $|\psi\rangle = (\hat{E} + i\lambda\hat{F})|\Phi\rangle$ with λ real for implicity, then:

$$\langle\psi,\psi\rangle \geq 0$$

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$$\langle \Phi | (\hat{E} - i\lambda\hat{F})(\hat{E} + i\lambda\hat{F}) | \Phi \rangle \geq 0.$$

$$\langle \Phi | E^2 + i\lambda\hat{E}\hat{F} - i\lambda\hat{F}\hat{E} + \lambda^2\hat{F}^2) | \Phi \rangle \geq 0.$$

$$\langle \Phi | \hat{E}^2 + \lambda^2\hat{F}^2 + i\lambda[\hat{E}, \hat{F}] | \Phi \rangle \geq 0$$

$$\langle \Phi | (\hat{A} - \langle\hat{A}\rangle)^2 + \lambda^2(\hat{B} - \langle\hat{B}\rangle)^2 + i\lambda[\hat{A} - \langle\hat{A}\rangle, \hat{B} - \langle\hat{B}\rangle] | \Phi \rangle \geq 0$$

$$\langle \Phi | (\Delta\hat{A})^2 + \lambda^2(\Delta\hat{B})^2 + i\lambda[\hat{A}, \hat{B}] | \Phi \rangle \geq 0 \rightarrow (1)$$

Minimum will occur, when:

$$\frac{d}{d\lambda}\left\{ \langle \Phi | (\Delta\hat{A})^2 + \lambda^2(\Delta\hat{B})^2 + i\lambda[\hat{A}, \hat{B}] | \Phi \rangle \right\} = 0.$$

$$2\lambda(\Delta\hat{B})^2 + i[\hat{A}, \hat{B}] | \Phi \rangle = 0.$$

$$2\lambda(\Delta\hat{B})^2 = -i\langle \Phi [\hat{A}, \hat{B}] | \Phi \rangle.$$

$$\lambda = \frac{-i\langle \Phi [\hat{A}, \hat{B}] | \Phi \rangle}{2(\Delta\hat{B})^2}$$

putting the value of λ in eq (1):-

$$\Rightarrow \langle \Phi | (\Delta\hat{A})^2 - \frac{|\langle \Phi [\hat{A}, \hat{B}] | \Phi \rangle|^2}{4(\Delta\hat{B})^4} \cdot (\Delta\hat{B})^2 + i\left(-\frac{i\langle \Phi | [\hat{A}, \hat{B}] | \Phi \rangle}{2(\Delta\hat{B})^2}\right) \geq 0$$

$$(\Delta\hat{A})^2 + \frac{|\langle \Phi | [\hat{A}, \hat{B}] | \Phi \rangle|^2}{4(\Delta\hat{B})^2} \geq 0.$$

$$(\Delta\hat{A})^2 \geq -\frac{|\langle \Phi | [A, B] | \Phi |^2}{4(\Delta\hat{B})^2}$$

$$(\Delta\hat{A})^2(\Delta\hat{B})^2 \geq -\frac{|\langle \Phi | [\hat{A}, \hat{B}] | \Phi |^2}{4}$$

$\because [\hat{A}, \hat{B}] = i\hat{C}$

$$(\Delta\hat{A})^2(\Delta\hat{B})^2 \geq \frac{-i^2}{4} |\langle \Phi | i\hat{C} | \Phi |^2$$

$$(\Delta\hat{A})^2(\Delta\hat{B})^2 \geq \frac{1}{4} |\langle \hat{C} \rangle|^2$$

$$\Rightarrow (\Delta\hat{A})(\Delta\hat{B}) \geq \frac{1}{2} |\hat{C}|$$

which completes the proof.

(II)

Differential Equation for SHO obtained after solving schrödinger wave equation is given by:

$$H''(z) - 2z\,H'(z) + (\lambda-1)H(z) = 0.$$

solve this eq. to find energy eigen value and function of simple Harmonic oscillator.

Since,

$$H''(z) - 2z\,H'(z) + (\lambda-1)H(z) = 0.$$

The solution of this equation is given as:

$$a_{n+2} = \frac{(2n-\lambda+1)}{(n+1)(n+2)}\,a_n \quad , n = 0,1,2,3,\ldots$$

This equation is called recursion formula.

To obtain finite values of wave function and energy eigen values, a_0 must be zero for odd "n" and a_1 must be zero for even n.

For some values of n:

$a_{n+2} = 0$

$$\Rightarrow \frac{(2n-\lambda+1)}{(n+1)(n+2)} a_n = 0 \Rightarrow (2n-\lambda+1) = 0$$

$$\lambda = (2n+1).$$

Energy Eigen values:-

Since,

$$\lambda = \frac{2E_n}{\hbar\omega} \rightarrow (i)$$

also, $\lambda = (2n+1) \rightarrow (ii)$

From eq(i) and (ii):-

$$\Rightarrow \frac{2E_n}{\hbar\omega} = (2n+1).$$

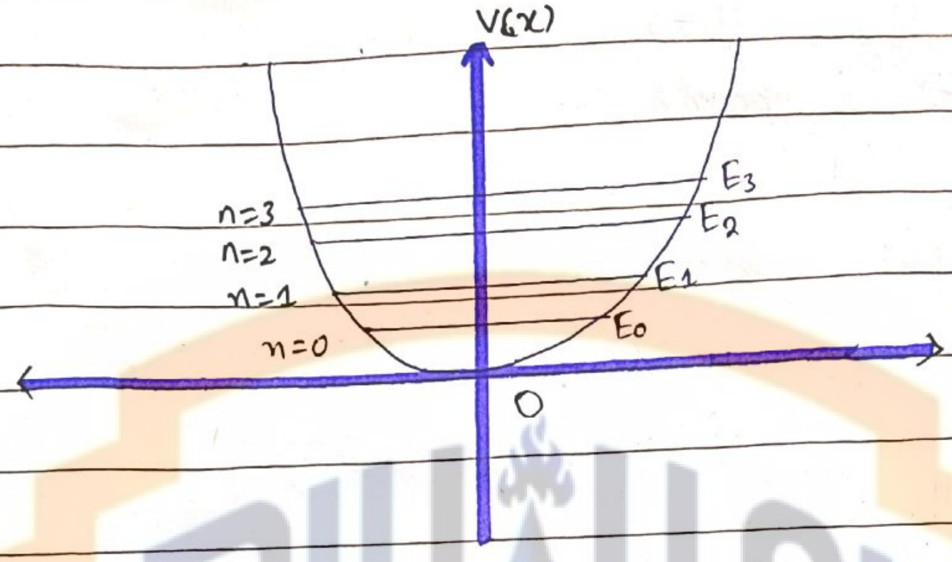
$$\Rightarrow E_n = \frac{(2n+1)\hbar\omega}{2}.$$

$$\Rightarrow E_n = (n+1/2)\hbar\omega.$$

For $n=0$, energy is given as:

$$E_0 = \frac{\hbar\omega}{2}$$

This energy is called zero point energy, which is characterstic of quantum mechanics- since energy of harmonic oscillator depends on "n" so all the energy levels of oscillator are non-degenrate. The successive energy levels are equally spaced and are shown in figure.



Wave functions:-

The wave function for harmonic oscillator for state En is given by:

$$\psi_n = a_n H_n(z) e^{-z^2/2}.$$

Here, "an" is normalization constant which can be calculated by orthogonality of Hermition polynomials.

$$a_n = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot \frac{1}{\sqrt{2^n \cdot n!}} \quad \text{where, } \alpha = \frac{m\omega}{\hbar}$$

so,

$$\psi_n = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot \frac{1}{\sqrt{2^n \cdot n!}} H_n(z) e^{-z^2/2}.$$

and $\quad z = x\sqrt{\alpha} \Rightarrow Z = x\sqrt{\frac{m\omega}{\hbar}}$

Hence, $\quad \psi_n(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot \frac{1}{\sqrt{2^n \cdot n!}} H_n(x\sqrt{\alpha})\, e^{-\alpha x^2/2}$.

Here, $H(z)$ are Hermite polynomials of degree "n".

For $n=0$, wave function is given as:

$$\Rightarrow \psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} H_0(x\sqrt{\alpha})\, e^{-\alpha x^2/2}.$$

This wave function is called ground state wave function.

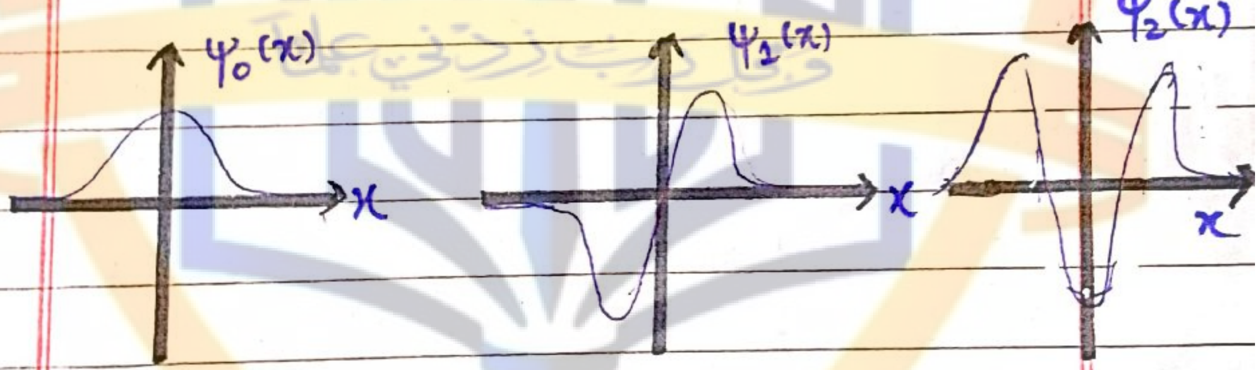


Figure: plot of three lowest energy states

For $n=1$, wave function is:

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot \frac{1}{\sqrt{2}} H_1(x\sqrt{\alpha})\, e^{-\alpha x^2/2}$$

$\because H_1 = 2$

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \cdot \frac{1}{\sqrt{2}} \cdot 2(x\sqrt{\alpha})\, e^{-\alpha x^2/2}$$

$$\psi_1(x) = \sqrt{2}\left(\frac{\alpha}{\pi}\right)^{1/4} \cdot (x\sqrt{\alpha})\, e^{-\alpha x^2/2}.$$

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From the above figure, it can be noted that even quantum numbers have symmetric wave functions $\psi_n(x)$ while odd quantum numbers have anti-symmetric wave functions $\psi_n(x)$.

Note:-

The wave function is either even or odd depending upon the value of "n". In fact $\psi_{2n}(x)$ function are even i.e.

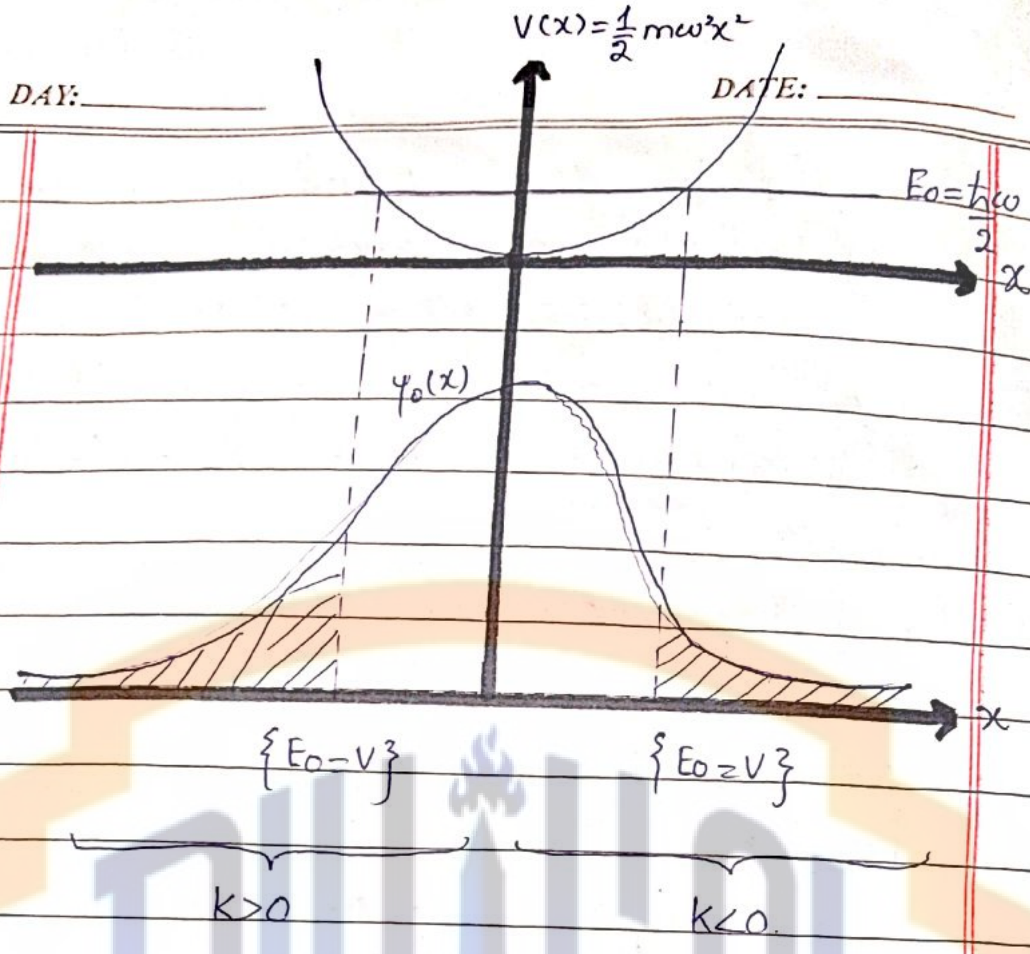
$$\psi_{2n}(-x) = \psi_{2n}(x)$$

and $\psi_{2n}(x)$ functions are odd. i.e.

$$\psi_{2n+1}(-x) = -\psi_{2n+1}(x).$$

This is due to the fact that Hermition polynomials $H_{2n}(x)$ are even and $H_{2n+1}(x)$ are odd.

Quantum mechanical description of energy and ground state wave function is plotted as following:



(III)

Discuss the significance of Laddar operator $\hat{L}_\pm$. show that they have following property:

$$L_+\psi(l,m)=\sqrt{l(l+1)-m(m+1)}\;\hbar\,\psi(l,m+1).$$

Proof:-

Eigen value equation for $\hat{L}_z$ and $\hat{L}^2$ are:

$$\hat{L}_z\psi(l,m)=m\hbar\,\psi(l,m)$$

$$\hat{L}^2\psi(l,m)=l(l+1)\hbar^2\psi(l,m).$$

operating first equation by $\hat{L}_+$:

$$\Rightarrow\quad \hat{L}_+\hat{L}_z\psi(l,m)=m\hbar\,\hat{L}_+\psi(l,m).$$

$$\Rightarrow (\hat{L}_z\hat{L}_+ - \hbar\hat{L}_+)\psi(l,m) = m\hbar\hat{L}_+\psi(l,m)$$

$$\hat{L}_z\hat{L}_+\psi(l,m) - \hbar\hat{L}_+\psi(l,m) = m\hbar\hat{L}_+\psi(l,m)$$

$$\Rightarrow \hat{L}_z\hat{L}_+\psi(l,m) = m\hbar\hat{L}_+\psi(l,m) + \hbar\hat{L}_+\psi(l,m)$$

$$\hat{L}_z\hat{L}_+\psi(l,m) = \cancel{m}\hbar\hat{L}_+(m+1)\psi(l,m)$$

$$\Rightarrow \hat{L}_z\hat{L}_+\psi(l,m) = (m+1)\hbar\hat{L}_+\psi(l,m)$$

Multiplying b.s. with C:-

$$\Rightarrow \hat{L}_z\{C\hat{L}_+\psi(l,m)\} = (m+1)\hbar\{C\hat{L}_+\psi(l,m)\} \quad (1)$$

Thus, $\{C\hat{L}_+\psi(l,m)\}$ is an eigen function of $\hat{L}_z$ corresponding to eigen value $(m+1)\hbar$. Now,

$$\hat{L}_+\hat{L}^2\psi(l,m) = l(l+1)\hbar^2\hat{L}_+\psi(l,m)$$

$$\hat{L}^2\hat{L}_+\psi(l,m) = l(l+1)\hbar^2\hat{L}_+\psi(l,m).$$

Multiplying both sides with "C":-

$$\Rightarrow \hat{L}^2\{C\hat{L}_+\psi(l,m)\} = l(l+1)\hbar^2\{C\hat{L}_+\psi(l,m)\} \rightarrow (2)$$

From eq (1) and (2), it can be seen that "l" remains constant / same but "m" increases by 1. So we can write:

$$C\hat{L}_+\psi(l,m) = \psi(l,m+1)$$

$$\Rightarrow \hat{L}_+\psi(l,m) = \frac{1}{C}\psi(l,m+1)$$

$$\hat{L}_+\psi(l,m) = A\psi(l,m+1) \rightarrow (3) \qquad \because \frac{1}{C} = A$$

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Taking complex conjugate:

$$\Rightarrow \{\hat{L}_+\psi(l,m)\}^* = A^*\psi^*(l,m+1) \rightarrow (4)$$

"x" eq (3) and (4) and integrating:-

$$\Rightarrow \int_{-\infty}^{+\infty} \{\hat{L}_+\psi(l,m)\}^* \hat{L}_+\{\psi(l,m)\}\,d\tau = \int_{-\infty}^{+\infty} A^*\psi^*(l,m+1)A.\,\psi(l,m+1)\,d\tau$$

$$\int_{-\infty}^{+\infty} \psi^*(l,m)\,\hat{L}_-\hat{L}_+\psi(l,m)\,d\tau = |A|^2\int_{-\infty}^{+\infty} \psi^*(l,m+1)\psi(l,m+1)\,d\tau$$

$$\int_{-\infty}^{+\infty} \psi^*(l,m)\,(\hat{L}^2-\hat{L}_z^2-\hbar\hat{L}_z)\,\psi(l,m)\,d\tau = |A|^2$$

$$|A|^2 = \int_{-\infty}^{+\infty} \psi^*(l,m)\{l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar^2)\,\psi(l,m)\,d\tau$$

$$|A|^2 = \{l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar^2)\int_{-\infty}^{+\infty} \psi^*(l,m)\,\psi(l,m)\,d\tau$$

$$\Rightarrow |A|^2 = (l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar^2\,)\,(1)$$

$$\Rightarrow A = \sqrt{l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar^2}$$

$$A = \sqrt{l(l+1)\hbar^2 - m(m+1)\hbar^2}$$

putting the value of "A" in eq(3):

$$\Rightarrow L_+\psi(l,m) = \sqrt{l(l+1) - m(m+1)}\;\hbar\,\psi(l,m+1)$$

$\Rightarrow$

Hence proved.